Abhay Bhandarkar

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Objective

Dynamic third-year Computer Science Engineering undergraduate researcher with a profound interest in applied Machine Learning, Deep Learning, and Computer Vision in Medical and Aerospace domains. Currently focused on cutting-edge architectures for medical imaging, AI guardrails, and social innovation. Possesses a strong foundation in theoretical engineering and extensive experience in transforming innovative ideas into tangible solutions. Skilled in collaborating with diverse teams and simplifying complex technical concepts for broader audiences.

Education

- **Ramaiah Institute of Technology** — *Bengaluru, Karnataka*
  Bachelor of Engineering in Computer Science (GPA: 7.4/10 [4th Semester]) — Expected Sept 2026

Publications

C=Conference, J=Journal, P=Patent, S=In Submission, W=In writing process

[P.1] **Abhay Bhandarkar**, et al. (In Submission). **JAM Algorithm for Probabilistic String Search**. Patent application submitted.

[P.2] **Abhay Bhandarkar**, et al. (In Submission). **SIMHA: Sustainable Integrated Microbial and Hydrogen fuel cell Architecture**. Patent application submitted.

[C.1] **Abhay Bhandarkar**, Dr. Nagaraju Kottam, Vishwachetan D (2023). **Minds and Materials in Flight: The Dynamic Duo of Smart Materials and Machine Learning in Aerospace Evolution**. International Conference for Advanced Materials in Engineering Sciences (ICAMES).

[C.2] **Abhay Bhandarkar**, Vishwachetan D (2023). **Unmasking the AI Hand: A Machine Learning Approach to Deciphering Authorship**. IEEE INOCON 2024.

[C.3] **Abhay Bhandarkar**, Riddhi Rai, Sushma Bylaiah. **Advancements in Defense Surveillance: Integrating Deep Learning and Enhanced Materials in CubeSAT Systems**. IEEE SPACE 2024.

[C.4] Manjula R Chougala, Sushma Bylaiah, **Abhay Bhandarkar**, Akshata S Bhayyar, P N Pavitra (2024). **TotCareBot: A Comprehensive Approach towards Enhancing Early Childhood Education through Robotics**. In *I4C 2024*.

[C.5] **Abhay Bhandarkar**, et al. (In Submission). **SIMHA: Sustainable Integrated Microbial and Hydrogen fuel cell Architecture**. IEEE SPACE 2025

[W.1] **Abhay Bhandarkar**, et al. (In writing process). **CAMMEL-ViT Net: Context Aware Memory Modules with Efficient Lightweight Vision Transformers for Lightweight Efficient Image Segmentation**.

[W.2] **Abhay Bhandarkar**, Khushi, Gaurav Mishra et al. (In writing process). **Topic Modelling with BERTopic: Understanding the likings of humans across various LLMs via thorough analysis of LMSYS dataset**.

[W.3] **Abhay Bhandarkar**, et al. (In writing process). **MITRA: Model In The Rear Approach to Safeguarding LLMs for low cost and scalable solutions**.

[W.4] **Abhay Bhandarkar**, et al. (In writing process). **Transformer+GNN, Transformer+LGM architectures for multi-class disaster classification over LADIV2 dataset**.

[W.5] **Abhay Bhandarkar**, et al. (In writing process). **A SriChakra based patch antenna for space applications**.

[W.6] **Abhay Bhandarkar**, et al. (In writing process). **Sanskrit Based Cipher with Dynamic Dictionary for Enhanced Security**.

Awards & Achievements

- Received $50000 from IBM in Compute Resources among all department students at RIT to carry out research in space sustainability
- Secured 1st place in CNS Hackathon as solo participant against entire CS department with MITRA AI Safety Framework
- Won 2nd prize in ACM Hackathon for CAMM-EL ViT Net medical imaging solution
- Achieved 2nd place among 800+ participants in Whackiest'24 by CodeRIT
- Qualified for Regionals in the Solving For India Hackathon by GFGXGoogleCloud

EXPERIENCE

Data Science Intern *2024*
Product Team Altqube, India

- Developed a YOLOv8 model for detection of Lumpy Skin Disease in cattle with 96% accuracy.
- Devised a pipeline for Oriented Bounding Box format to analyze rice grain quality in an industrial setting.
- Implemented a pipeline for training medical images with Nomic AI Vision Embedding model.
- Annotated 600+ images with multi-label bounding boxes on Roboflow to facilitate robust YOLOv8 training.

Project Intern *2023*
Laboratory for Electro-Optics Systems Indian Space Research Organization (ISRO) LEOS, Bangalore, India

- Contributed to LiDAR point cloud analysis for an autonomous docking project (classified).

Machine Learning Intern *2023*
Government Transliteration Project Crisant Technologies Karnataka, India

- Assisted with transliteration from English to Kannada for streamlined documentation for the Karnataka government.
- Fine-tuned Google-MT5 small model for improved transliteration via transformer-based approaches.

RESEARCH PROJECTS

- **CAMM-EL ViT Net** *2025*
  Python, PyTorch, U-Net, Transformers
  - **Description:** Proposed a novel deep learning architecture: **Context-Aware Memory Modules and Efficient Lightweight Transformers** with attention-guided connections for enhanced medical image segmentation. Integrates a U-Net backbone with lightweight transformer modules and context-aware memory for superior feature extraction and spatial understanding.
  - **Performance Highlights:**
    - **BraTS2020 Dataset Dice Coefficients:** ET = 0.955 (Train), 0.9198 (Val); NEC = 0.9676 (Train), 0.9486 (Val); WT = 0.9996 (Train), 0.9994 (Val).
    - **IoU Metrics:** ET = 0.929 (Train), 0.8854 (Val); NEC = 0.9513 (Train), 0.9279 (Val); WT = 0.9992 (Train), 0.9987 (Val).
    - Achieved state-of-the-art performance with a lightweight 260.65 MB model, trained in just 29 epochs.
    - Very fast inference time of 127 ms on average.
  - **Impact:** Highly effective in brain tumor segmentation tasks. Also deployed using Streamlit locally with the vision of reaching poor resourced areas around the world in an attempt to aid radiologists and neurologists with better tumour segmentation.
  - **Status:** Publication incoming soon, applying to ICCV 2025.
- **MITRA: Model in the Rear Approach to Safeguarding LLMs**
  Python, Flask, Hugging Face
  - **Purpose:** A novel **LLM Guardrail** method using classification LLMs to protect the main LLM from jailbreaks and other harmful prompts.
  - **Key Features:**
    - Multi-layer Safety Checks (toxicity detection, jailbreak detection).
    - Embedding-based threat analysis and real-time classification.
    - Auto-resizing, responsive chat interface with confetti feedback.
    - Rate limiting to prevent abuse and rotating example prompts.
  - **Technologies and Models:**
    - **Hugging Face Transformers**: RoBERTa-based toxicity classifier; Jailbreak Detector (madhurjindal/Jailbreak-Detector-Large).
    - **SentenceTransformers**: all-MiniLM-L6-v2 for semantic embeddings.
    - **Qwen2.5:0.5b** via LangChain Ollama for the core LLM.
  - **Status:** Publication incoming soon, detailing methodology, architecture, and performance results.
- **Sanskrit Cipher System: Secure Messaging with Dynamic Dictionary Mapping** *2024*
  Python, Secrets, String
  - **Overview:** A dynamic encryption and decryption system using Sanskrit syllables as cipher elements. This system maps each letter of the English alphabet to a unique Sanskrit syllable and utilizes a Caesar cipher approach for secure message transmission.

- **Key Features:**
  - **Dynamic Dictionary Generation**: Uses Sanskrit syllables for encryption based on random mapping of English letters to unique Sanskrit characters.
  - **Caesar Cipher Implementation**: Encrypts messages by shifting the letters in the English alphabet and mapping the result to Sanskrit characters.
  - **Decryption Process**: Uses a reverse process for decryption, ensuring that the original message is restored accurately.
  - **Support for Non-Alphabetic Characters**: The system handles non-alphabetic characters such as punctuation and spaces, which are retained in the ciphertext.
- **Technologies Used:**
  - **Python**: Core programming language for encryption and decryption logic.
  - **Secrets and String Modules**: Used for secure randomization of the Sanskrit syllables and manipulation of text.
- **Impact:** The project provides a secure method for communication using a novel Sanskrit-based encryption technique, ideal for educational applications and promoting secure messaging practices in various languages.
- **Status:** Active, currently refining the encryption system and preparing for deployment in educational settings.

Leadership Experience

- **Project Manager, S.T.A.R.D.U.ST – RIT** *2023 - August 2023*
  Bangalore, India
  - Contributed to flight computer design for Spaceport America Cup 2025.
- **Founder, IEEE Student Team for Research & Innovation (ISTRI)** *2024 - Present*
  RIT, Bangalore, India
  - Spearheaded a platform for real-world problem-solving research collaborations with IEEE-RIT.
  - Organized student-led conferences to promote innovation and practical engineering.
- **Chair, IEEE Computer Society - RIT** *2024 - Present*
  Bangalore, India
  - Organized technical events and hackathons to bolster student engagement in computer science.

Skills

- **Languages:** English, Hindi, Kannada, Konkani, Sanskrit
- **Programming:** Python, Julia (basics), C++, C, HTML, CSS, React.js
- **Frameworks/Tools:** PyTorch, Google Colab, VSCode, Flutter, Firebase, Flask

Relevant Courses

Linear Algebra, Machine Learning, Design and Analysis of Algorithms

Certifications

- NPTEL - IIT Kharagpur: Deep Learning
- Google: Create and Manage Cloud Resources
- LinkedIn: Python Quick Start
- LinkedIn: Web Programming Foundations

References

- **Harsh Singhal** Adobe Engineer
- **Jamuna S Murthy** Assistant Professor, RIT